

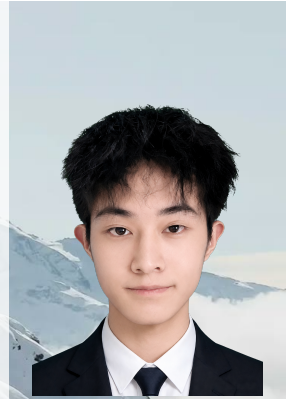
# Yongxue Xu (Jerry)

Undergraduate researcher at Sun Yat-sen University

*Video generation • World models • 4D scene understanding*

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## Profile

Undergraduate researcher working on video understanding, object-centric temporal modeling, and 4D dynamic reasoning. Seeking an industry research internship in multimodal foundation models, video generation, or world models.

Available from October 2026 for 6+ months, through graduation in June 2027.

## Education & Toolkit

### Sun Yat-sen University

SEP 2023 -- JUN 2027

B.Eng. in Intelligent Science and Technology

**GPA 3.91 / 4.0 • Major GPA 3.94 / 4.0**

Project 985 / 211 / Double First-Class

### RESEARCH

Video understanding/generation, multimodal foundation models, world models, 4D scenes, WAM.

### TOOLS & LANGUAGES

Python, C/C++, MATLAB, PyTorch; Chinese (native), English (CET-6).

## Research Experience

### LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

*Co-first author / Preprint*

Developed [LMM-Track4D](#) for multi-view 4D reasoning with geometry-aware encoding, persistent state memory, and trajectory decoding. Co-developed Track4D-Bench with 526 dialogues, 23.5k frames, and 7.5k object annotations.

Achieved substantial gains over the base model on external benchmarks, particularly in dynamics reasoning, alongside 61.9% trajectory-step accuracy within 0.5 m on Track4D-Bench.

**Contribution:** Led model design and main experiments; co-developed the benchmark.

### Learning to Reason with Persistent Object States for Video Instance Segmentation

*Co-first author / Preprint*

Developed [POSReasoner](#) for long-horizon identity reasoning, coupling persistent object memory with verified state transitions to counter identity drift and memory corruption under occlusion and reappearance.

Enhanced object recovery on LVOS v1: +13.1  $\mathcal{J}$  after 11–30 absent annotated frames and +3.5 J&F overall, without fine-tuning SAM3.

**Contribution:** Led method and architecture design, implementation, experiments, and manuscript preparation.

### Graph Domain Adaptation Does Not End with Representation Learning

*Co-first author / Preprint*

Developed [EviGDA](#): independently trained graph-aware and feature-only predictors, entropy-aware domain alignment, and fixed-weight probability fusion without learned routing or target pseudo-labels.

Achieved state-of-the-art average performance across 16 transfer tasks on 10 datasets: 67.67 Macro-F1 over five runs.

**Contribution:** Designed the core method and led implementation, evaluation, and manuscript development.

### Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach

*First author / TNSE  
under review*

Extended robust competitive influence maximization (RCIM) from single-layer to multilayer networks with intra-layer competition, cross-layer collaboration, and node-removal robustness. Developed [DRCIM-ML](#), using diffusion-aware role alignment and evolutionary search to transfer seed candidates across layers.

**Contribution:** Led problem formulation, method design, experiments, and writing; one patent pending.

## Selected Publications

**LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue** 2026

Chaoyue Li\*, Yongxue Xu\*, Jie Feng\*, and Jiayu Ding†

Preprint; [arXiv:2605.19390](#)

**Learning to Reason with Persistent Object States for Video Instance Segmentation** 2026

Yongxue Xu\*, Boxue Yang\*, Ziqian Liu, Shaoqiu Zhang, and Linfeng Zhang†

Preprint

**Graph Domain Adaptation Does Not End with Representation Learning** 2026

Yongxue Xu\*, Ziqian Liu\*, Enze Zhang, and Maolin Wang†

Preprint

**Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach** 2026

Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang†

Under review at *IEEE Transactions on Network Science and Engineering*

**Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks** 2025

Zhonghan Liu, Wenshuo Liu, Yongxue Xu, Ziqian Liu, Yichen Liu, and Shuai Wang†

GBCESC 2025 / Best Paper Award

\* Equal contribution. † Corresponding author.

## Research Initiatives & Collaborations

Collaborate with researchers at **HKUST**, **SJTU**, and **Westlake University** on video generation and multimodal spatiotemporal understanding. Together with peers in **InkMind.AI**, explore multimodal understanding and generation and refine AI-assisted research workflows through hands-on projects.

## Additional Research

**Urban road-network analysis** (GBCESC 2025, Best Paper): contributed dataset construction, GNN/DQN baseline reproduction, and experimental visualization for critical-node identification.

## Awards & Competitions

|                                                                                 |           |
|---------------------------------------------------------------------------------|-----------|
| Sun Yat-sen University First-Class Scholarship                                  | 2023–2026 |
| Chinese Mathematics Competition (CMC), Guangdong Region – Second Prize          | Dec. 2024 |
| Mathematical Contest in Modeling (MCM), USA – Honorable Mention                 | Feb. 2025 |
| National Undergraduate Smart Car Competition, South China Region – Second Prize | Aug. 2025 |

## Beyond Research

Table tennis, music, and nature travel.